

09-Observability-Architecture.md

Healthcare Clinical Intelligence Platform

Enterprise Observability Architecture

Version: 1.0

Status: Approved Design

Mission

Provide complete visibility into:

- AI Agents
- Retrieval Systems
- LLM Usage
- User Activity
- Infrastructure
- Costs
- Governance
- Compliance

The goal is to know exactly:

What happened?
Why did it happen?
How much did it cost?
How can it be improved?

Observability Philosophy

Bad:

User reports issue
↓
Start debugging

Good:

```
Issue Occurs
↓
System Detects
↓
System Alerts
↓
Root Cause Available
```

Observability Pillars

Logs

What happened?

Metrics

How often?

Traces

What path was executed?

Events

Why did it happen?

Observability Stack

Langfuse

AI Observability

OpenTelemetry

Distributed Tracing

Prometheus

Metrics Collection

Grafana

Dashboards

Loki

Application Logs

System Architecture

```
graph TD; Frontend --> FastAPI; FastAPI --> LangGraph; LangGraph --> Retrieval_Layer[Retrieval Layer]; Retrieval_Layer --> Models;
```

Every step emits:

```
Logs
Metrics
Traces
Events
```

Trace Architecture

Every request gets:

trace_id

Example:

```
graph TD
    A[User Query] --> B[Planner Agent]
    B --> C[Retriever]
    C --> D[Validator]
    D --> E[Safety Agent]
    E --> F[Response Agent]
```

Entire path visible.

Trace Hierarchy

```
graph TD
    A[Request Trace] --> B[Agent Span]
    B --> C[Tool Span]
    C --> D[LLM Span]
```

Agent Observability

Track:

Planner Decisions

Retriever Decisions

Validation Results

Safety Decisions

Response Generation

Example

Planner Selected:
Hybrid Retrieval

Reason:
Clinical Query

Retrieval Observability

Track:

Retrieved Chunks

Sources Used

Rerank Scores

Confidence Scores

Retries

GraphRAG Usage

Example

Chunks Retrieved: 78

Chunks Used: 12

Source Diversity: 4

Confidence: 0.92

LLM Observability

Track:

Model

Prompt Version

Input Tokens

Output Tokens

Latency

Cost

Errors

Example

Model:
Claude Sonnet

Input:
4200 Tokens

Output:
850 Tokens

Cost:
\$0.02

Prompt Observability

Track:

Prompt Version

Prompt Changes

Performance

Cost Impact

Quality Impact

Memory Observability

Track:

Memory Reads

Memory Writes

Memory Hits

Memory Misses

Memory Quality

Knowledge Base Observability

Track:

Document Uploads

Document Approval

Document Rejections

Version Updates

Indexing Status

Ingestion Observability

Track:

Upload Latency

Parse Latency

OCR Latency

Chunk Count

Embedding Cost

Graph Nodes Created

Failures

Evaluation Observability

Track:

Context Precision

Context Recall

Faithfulness

Hallucinations

Citation Coverage

Safety Accuracy

User Analytics

Track:

Active Users

Queries Per User

Departments

Top Knowledge Bases

Popular Questions

Business Metrics

Track:

Organizations

Daily Active Users

Monthly Active Users

Query Volume

Revenue

Cost

Profitability

Cost Observability

Track:

Cost Per Query

Cost Per Agent

Cost Per Model

Cost Per Organization

Cost Per Knowledge Base

Cost Breakdown

Retrieval

Embeddings

LLM

Storage

Infrastructure

Governance Observability

Track:

RBAC Violations

Access Denials

Approval Events

Policy Violations

Audit Events

Security Monitoring

Track:

Failed Logins

Suspicious Activity

API Abuse

Unauthorized Access

Tenant Violations

Alerting System

Critical Alerts

High Hallucination Rate

>5%

Retrieval Failure

>10%

Agent Failure

>5%

Cost Spike

2x normal cost

Latency Spike

>10 seconds

SLA Monitoring

Target:

99.9% Availability

SLO Monitoring

Response Time:

<5 seconds

Retrieval:

<1.5 seconds

Agent Execution:

<3 seconds

Dashboard Structure

Executive Dashboard

Business Metrics

Operations Dashboard

Infrastructure Health

AI Dashboard

Agents

Retrieval

LLMs

Evaluation Dashboard

Quality Metrics

Security Dashboard

Compliance

Audit

Access

Cost Dashboard

Spend

Optimization

Savings

Incident Response

Every failure generates:

```
graph TD; Incident --> RootCause[Root Cause]; RootCause --> Alert; Alert --> Resolution;
```

Audit Trail

Every action logged:

User

Agent

Document

Retrieval

Approval

Memory

Administration

Data Retention

Logs

Metrics

Traces

Audit Events

All configurable by policy.

Future Observability

Agent-to-Agent Communication Monitoring

Autonomous Workflow Monitoring

Predictive Failure Detection

Cost Forecasting

Anomaly Detection

Success Targets

99.9% Availability

<5s Response Time

<1% Failed Queries

100% Audit Coverage

100% Trace Coverage

Final Principle

If we cannot observe it, we cannot trust it.

Every decision, every retrieval, every cost, every answer must be traceable.